

Practice Lab:

DSE Analytics, Join, nulls, .isDefined, .get, map, filter, .saveToCassandra, and more

- This Practice Lab is dependent on Discussion Unit 7570, where most of the objects we create in this lab were introduced.
- This Practice Lab has as a prerequisite Practice Lab 7445, where we loaded the DSE tables customer, orders, items from CSV.
- This Practice Lab requires a working DSE system, with DSE Analytics enabled.
- Because of the dependency on the "dse spark" utility, this Practice Lab requires a ssh(C) prompt on at least one node operating DSE Analytics.
- Or, compile and run programs using an IDE.

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To: DSE_HOTSHOT

Subject: Need this now !!!

Big deal. You previously loaded 3 tables. ☹

Now join the SQL modeled customer and orders, and load the DSE modeled customer_orders.

I also need Jobs for:

- . Customers without orders
- . Count of orders by state
- . Orders where paid_date is null
- . Count of orders by customer, sort by count descending

-MA

Working Goal:



Maury Atwater, President
of Atwater's

```
DROP KEYSPACE IF EXISTS ks_7571;
```

```
CREATE KEYSPACE ks_7571 WITH REPLICATION =  
    {'class': 'SimpleStrategy',  
     'replication_factor': 1};  
USE ks_7571;
```

```
CREATE TABLE customer  
(  
    customer_num          INT PRIMARY KEY,  
    fname                 TEXT,  
    lname                 TEXT,  
    company               TEXT,  
    address1              TEXT,  
    address2              TEXT,  
    city                  TEXT,  
    state                 TEXT,  
    zipcode               TEXT,  
    phone                 TEXT  
);
```

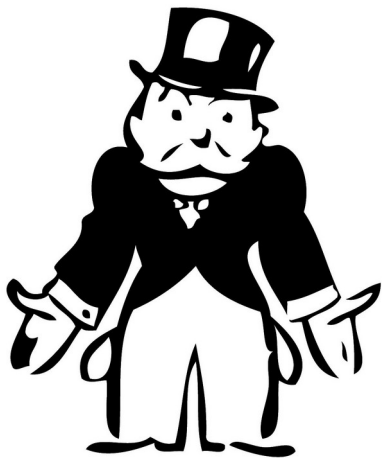
CQL Assets for Reference:

```
CREATE TABLE orders
(
  order_num          INT PRIMARY KEY,
  order_date         TEXT,
  customer_num       INT,
  ship_instruct      TEXT,
  backlog            TEXT,
  po_num             TEXT,
  ship_date          TEXT,
  ship_weight        FLOAT,
  ship_charge        FLOAT,
  paid_date          TEXT
);
```

```
CREATE TABLE customer_orders
(
  order_num          INT,
  paid_date          TEXT,
  customer_num       INT,
  state              TEXT,
  PRIMARY KEY((order_num), customer_num)
);
```

CQL Assets for Reference:

Challenge 1: Customers with no Orders



Source: <https://www.pinterest.com/pin/5629568257464509/>

- Read from customer and orders
- Use a leftOuterJoin
- Then filter using .isDefined
- Just read; customer_num (both tables), state, order_num, paid_date
- Why ? We can re-use these RDDs later
- There are 11 rows by count

Challenge 2: Load customer_orders

By **far** the hardest of the challenges

- It's always the little things; nulls, nits in the APIs, yadda
- Keep it simple; we're only keeping 4 columns to reduce burden
- Best effort, then check answer at end of unit
- You need customer_orders for the remaining challenges
- All other challenges are do-able



Challenge 3: Count of Orders by State

- Need customer_orders
- 4 lines; map, and reduceByKey()
- Output just state and count



Source: <https://www.eater.com/2016/9/26/13051528/most-ordered-food-delivery-america>

Challenge 4: Rows where paid_date is (null)



- Mismatch between: DSE, Spark, Scala
- Advice in Scala is to remove nulls immediately; use a constant
- Spark has methods
- DSE has other ideas of its own
- Our solution used a constant, ("N/A")

Challenge 5: High Value Customers

- Count of orders by customer, sort by count descending
- Output customer number and count
- Bonus: also output state
- 5 lines



Practice Lab:

Lessons learned





End of Unit:



Additional Detail:

Answer to just challenge 2 is on the Notes page-

